Inference for linear models

→ Model $Y = X\beta + \varepsilon$ $\quad p$ covariates $X_{n\times(p+1)}$ $\quad \varepsilon \sim N_n(0, \sigma^2 I)$

→ Likelihood $L(\beta, \sigma^2 \mid Y) = \dfrac{1}{(2\pi)^{n/2}\sigma^n} \exp\left\{-\dfrac{1}{2\sigma^2}(Y - X\beta)^T(Y - X\beta)\right\}$

$$\ell(\quad) = -\frac{n}{2}\ln 2\pi - n\ln\sigma - \frac{1}{2\sigma^2}\underline{(Y - X\beta)^T(Y - X\beta)},$$

Max ℓ w.r.t $\beta \iff$ min sum of squares.

$$\downarrow \hat\beta_{MLE} \qquad \qquad \downarrow \hat\beta_{LS}$$

→ MLE
$$\begin{cases} \hat\beta = (X^TX)^{-1}X^TY. \\ \hat\sigma^2 = \dfrac{1}{n}(Y - X\hat\beta)^T(Y - X\hat\beta) = \dfrac{1}{n}SSR. \end{cases}$$

$$\rightarrow \begin{cases} \hat\beta = \beta + (X^TX)^{-1}X^T\varepsilon \\ \hat\beta \sim N(\beta, \sigma^2(X^TX)^{-1}) \end{cases}$$

$\varepsilon \sim N(0, \sigma^2 I)$
$A = (X^TX)^{-1}X^T$
$b = \beta$

1. $X \sim N(\mu, \Sigma)$
   $AX + b \sim N(A\mu + b, A\Sigma A^T)$
2. $X \sim N(0, \Sigma)$
   Σ is idempotent ($\Sigma^2 = \Sigma$)
   then
   $X^TX \sim \chi^2_k$ where $k = \text{rank}(\Sigma)$

$$Y - X\hat\beta = [I - X(X^TX)^{-1}X^T]\varepsilon$$

$$\underbrace{Y - X\hat\beta}_{} \sim N(0, \sigma^2\underbrace{(I - X(X^TX)^{-1}X^T)}_{})$$

$$\frac{n\hat\sigma^2}{\sigma^2} = \left[\frac{1}{\sigma}(Y - X\hat\beta)\right]^T\left[\frac{1}{\sigma}(Y - X\hat\beta)\right] \sim \chi^2_k$$

$$k = \text{rank}[I - X(X^TX)^{-1}X^T]$$
$$= n - (p+1)$$

→ $Y - X\hat\beta = [I - X(X^TX)^{-1}X^T]\varepsilon$ and $\dfrac{n\hat\sigma^2}{\sigma^2} \sim \chi^2_{n-(p+1)}$

<u>To show</u>: $\text{rank}\{I - X(X^TX)^{-1}X^T\} = n - (p+1)$.

<u>Pf</u>: <u>Result</u> For A symmetric & idempotent, ($A^T = A$ and $A^2 = A$)
$\text{rank}(A) = \text{tr}(A)$.

$$\text{tr}(I - X(X^TX)^{-1}X^T) = \text{tr}(I) - \text{tr}(X(X^TX)^{-1}X^T)$$
$$= n - \text{tr}(X^TX(X^TX)^{-1})$$
$$= n - \text{tr}(I_{p+1}) = n - (p+1).$$

Proof of result: $A = V\Lambda V^{-1}$ spectral decomposition

Λ is diagonal matrix of eigenvalues of A.

$tr(A)$ = sum of eigenvalues of A $\quad \begin{bmatrix} 1 & 0 & \\ & \ddots & \\ 0 & & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 0 \end{bmatrix}$

$= \ldots \Lambda$

For idempotent matrix, eigenvalues are 0 or 1

$$\lambda x = \underset{=}{A} x = \underset{=}{A}^2 x = A(\lambda x) = \lambda(Ax) = \lambda.\lambda x = \lambda^2 x$$

$$(\lambda - \lambda^2)x = 0, \quad x \neq 0 \Rightarrow \lambda^2 = \lambda \Rightarrow \lambda = 0 \text{ or } 1$$

Sum of eigenvalues of Λ = # of 1's in the diagonal of Λ

= rank of Λ

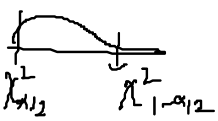
= rank of A

$$\frac{n\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-(p+1)} \qquad E\chi^2_k = k$$

$$E\hat{\sigma}^2 = \frac{\sigma^2}{n}(n-(p+1)), \quad \hat{\sigma}^2 \text{ is biased estimator for } \sigma^2.$$

$$E\left[\frac{n}{n-(p+1)}\hat{\sigma}^2\right] = \sigma^2. \qquad \frac{n\,\hat{\sigma}^2}{n-(p+1)} = \frac{SSRes}{n-(p+1)} := s^2 \text{ is unbiased for } \sigma^2$$

$\rightarrow$ CI for σ^2. $P\left[\frac{(n-(p+1))s^2}{\chi^2_{1-\alpha/2,\, n-(p+1)}} < \sigma^2 < \frac{(n-(p+1))s^2}{\chi^2_{\alpha/2,\, n-(p+1)}}\right] = 1-\alpha$



$\rightarrow$ CI for β

CI for β_i $\qquad \hat{\beta} \sim N(\beta, \sigma^2(X^TX)^{-1})$

σ^2 is unknown. s^2 $\quad$ indep $\quad \frac{N}{\sqrt{\chi^2/k}} \sim t$

- $\hat{\beta}$ and s^2 are indep.
- $p=1$ case reduces to simple linear regression
- ANOVA regression is significant
- CI for expected value of Y given $X = x_0$

  $E(Y|X=x_0) = x_0\hat{\beta}$
- CI for predicted value of Y.

To show $\hat{\beta}$ and s^2 are indep $\qquad Y = X\beta + \varepsilon$

To show $\hat{\beta}$ and s^2 are indep

$$\hat{\beta} = (X^TX)^{-1}X^TY \qquad s^2 = \frac{1}{n-(p+1)}(Y-X\hat{\beta})^T(Y-X\hat{\beta}).$$

$Y = X\beta + \varepsilon$

$\varepsilon \sim N(0, \sigma^2 I)$

To show $\hat{\beta}$ and $Y - X\hat{\beta}$ are indep.

$X \perp\!\!\!\perp Y$

$f(X) \perp\!\!\!\perp g(Y)$

$\rightarrow \hat{\beta} = \beta + (X^TX)^{-1}X^T\varepsilon$

$\rightarrow Y - X\hat{\beta} = [I - X(X^TX)^{-1}X^T]\varepsilon$

$\cancel{\hat{\beta} \text{ and } Y - X\hat{\beta}} \Rightarrow \begin{bmatrix} \hat{\beta} \\ Y - X\hat{\beta} \end{bmatrix} = \begin{bmatrix} \beta \\ 0 \end{bmatrix} + \begin{bmatrix} (X^TX)^{-1}X^T \\ I - X(X^TX)^{-1}X^T \end{bmatrix}\varepsilon$

$\varepsilon \sim N$

$b + A\varepsilon \sim N$

$\hat{\beta} \perp\!\!\!\perp (Y - X\hat{\beta})$ iff $\mathrm{cov}(\hat{\beta}, Y - X\hat{\beta}) = 0.$

For multivariate normal $\mathrm{cov} = 0 \Rightarrow$ indep

In general $\mathrm{cov} = 0$ $\not\Rightarrow$ indep

To show $\mathrm{cov}(\hat{\beta}, Y - X\hat{\beta}) = 0.$

$\varepsilon \sim N(0, \Sigma)$

$\mathrm{cov}(A\varepsilon, B\varepsilon) = A\Sigma B^T$

$$\begin{aligned}
\mathrm{cov}(\hat{\beta}, Y - X\hat{\beta}) &= \mathrm{cov}(\beta + (X^TX)^{-1}X^T\varepsilon, [I - X(X^TX)^{-1}X^T]\varepsilon) \\
&= \mathrm{cov}((X^TX)^{-1}X^T\varepsilon, [I - X(X^TX)^{-1}X^T]\varepsilon) \\
&= \sigma^2 (X^TX)^{-1}X^T[I - X(X^TX)^{-1}X^T] \\
&= \sigma^2[(X^TX)^{-1}X^T - (X^TX)^{-1}X^TX(X^TX)^{-1}X^T] = \sigma^2 \cdot 0 = 0.
\end{aligned}$$

Confidence interval for β_i

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$$

$\hat{\beta} \sim N_{p+1}(\beta, (X^TX)^{-1}\sigma^2)$

$\hat{\beta}_i \sim N(\boxed{\beta_i}, \sigma^2 (X^TX)^{-1}_{ii})$

CI for β_i : $\hat{\beta}_i \pm z_{\alpha/2} \cdot \sigma\sqrt{(X^TX)^{-1}_{ii}}$

$$\frac{\hat{\beta}_i - \beta_i}{s\sqrt{(X^TX)^{-1}_{ii}}} \sim t_{n-(p+1)}$$

$\hat{\beta}_i \perp\!\!\!\perp s \qquad \frac{\boxed{s^2(n-(p+1))}}{\sigma^2} \sim \chi^2_{n-(p+1)}$

$$\beta_i \in \hat{\beta}_i \pm t_{n-(p+1), \alpha/2}\, s\sqrt{(X^TX)^{-1}_{ii}}$$

$X_1 \ldots X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

$\bar{X} \sim N(\boxed{\mu}, \sigma^2/n)$

σ known

CI for μ: $\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\sigma/\sqrt{n}\right)$

σ unknown

$\bar{X} \pm t_{n-1, \alpha/2}\, s/\sqrt{n}$

$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t \qquad s = \frac{1}{n}\sum(X_i - \bar{X})^2$

$\bar{X} \perp\!\!\!\perp s$

$\cancel{\frac{s^2(n-p+1)}{\sigma^2} \sim \chi^2}$

$p = 1$. Simple linear regression.

$X = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & \end{bmatrix} \qquad s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$

$p=1$. Simple linear regression.

$X = \begin{bmatrix} \vdots & \vdots \\ 1 & x_n \end{bmatrix}$ $\quad s^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2$ $\quad Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$

$X^T X = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}_{2\times 2}$ $\quad (X^TX)^{-1} = \frac{1}{n\sum x_i^2 - (\sum x_i)^2} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}_{2\times 2}$

$n\sum x_i^2 - (\sum x_i)^2$
$= n^2\left[\frac{1}{n}\sum x_i^2 - \bar{x}^2\right]$
$= n^2 \frac{1}{n}\sum(x_i-\bar{x})^2$
$= \cancel{n(n-1)s^2} = nSS_{xx}$

$\hat{\beta} = (X^TX)^{-1}X^TY$ $\quad \hat{\beta} \sim N(\beta, \sigma^2(X^TX)^{-1})$

$X^TY = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}_{2\times 1}$

$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \hat{\beta} = \frac{1}{n(n-1)s^2} \begin{bmatrix} \sum x_i^2 \sum y_i - \sum x_i y_i \sum x_i \\ -\sum x_i \sum y_i + n\sum x_i y_i \end{bmatrix}$

$n\sum x_i y_i - \sum x_i \sum y_i$
$= n\sum(x_i-\bar{x})(y_i-\bar{y}) = nSS_{xy}$

$\hat{\beta}_1 = \frac{nSS_{xy}}{nSS_{xx}} = \frac{SS_{xy}}{SS_{xx}}$

Verify that $\hat{\beta}_0$ you get from here is same as LS estimates from Simple Linreg.

$\hat{\beta} \sim N\left(\beta, \sigma^2 \cdot \frac{1}{nSS_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}\right)$

~~CI for $\hat{\beta}_1$~~ $\quad \hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2 n}{nSS_{xx}}\right)$; $\frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{n-2}$

$\frac{N}{\sqrt{\chi^2_k/k}} \sim t$. $\quad \frac{\hat{\beta}_1 - \beta_1}{s/\sqrt{SS_{xx}}} \sim t_{n-2}$

CI for β_0, β_1

$\hat{\beta}_0 \sim N\left(\beta_0, \frac{\sigma^2 \sum x_i^2}{nSS_{xx}}\right)$

$\left(\hat{\beta}_0 \pm t_{n-2,\alpha/2} \cdot s\sqrt{\frac{\sum x_i^2}{nSS_{xx}}}\right)$

Hypothesis test for β_1 $\quad H_0: \beta_1 = b$ vs. $\beta_1 \neq b$.

$t = \frac{\hat{\beta}_1 - b}{s/\sqrt{SS_{xx}}}$ $\quad$ Reject H_0 if $|t| > t_{n-2,\alpha/2}$.

- $\hat{\beta}_i$ and s^2 are indep.
- $\frac{\hat{\beta}_i - \beta_i}{s\sqrt{(X^TX)^{-1}_{ii}}} \sim t_{n-(p+1)}$

- $p=1$ Obtained CI and hypothesis tests for β_0 & β_1

$\hat{y}_i = (X\hat{\beta})_i$ → predicted value of y

ANOVA

Source	SS	df	MS	
~~SS~~ Regres	SST − SSRes Q_3	p	SSReg/p	MSReg
~~SS~~ Resid	$\sum(y_i-\hat{y}_i)^2$ Q_2	$n-(p+1)$	SSRes/$(n-(p+1))$	$\frac{MSReg}{MSRes} \sim F_{p,n-p+1}$
SST	$\sum(y_i-\bar{y})^2$	$n-1$		

$$\sum_{i=1}^{n}(y_i-\hat{y}_i)^2 = \sum_{i=1}^{n}\hat{\varepsilon}_i = \hat{\varepsilon}^T\hat{\varepsilon} = (Y-X\hat{\beta})^T(Y-X\hat{\beta}) = (n-(p+1))\hat{\sigma}^2$$

Fisher Cochran Thm

If $\varepsilon \sim N_n(0,\sigma^2 I)$, symmetric matrices $B^{(1)}, B^{(2)} \dots B^{(k)}$ satisfying $\sum_{i=1}^{k} B^{(i)} = I$, $Q_i = \varepsilon^T B^{(i)}\varepsilon$, and $r_i = \text{rank}(B^{(i)})$

then the following are equivalent

(i) $\sum r_i = n$ ✓

(ii) $Q_i \sim \chi^2_{r_i}$

(iii) $Q_1, Q_2, \dots Q_k$ are indep.

$$\hat{\varepsilon} = Y - X\hat{\beta} = X\beta+\varepsilon - X(X^TX)^{-1}X^T(X\beta+\varepsilon)$$
$$= X\beta+\varepsilon - X\beta - X(X^TX)^{-1}X^T\varepsilon$$
$$= [I - X(X^TX)^{-1}X^T]\varepsilon$$

$$SSRes = \sum(y_i-\hat{y}_i)^2 = (Y-X\hat{\beta})^T(Y-X\hat{\beta})$$
$$= \varepsilon^T[I - X(X^TX)^{-1}X^T]\varepsilon$$

$\frac{1}{n}1'Y = \bar{y}$ $\quad 1 = \begin{pmatrix}1\\ \vdots \\ 1\end{pmatrix}$

$$SSTotal = \sum(y_i-\bar{y})^2$$
$$= \sum y_i^2 - n\bar{y}^2$$

$\varepsilon^T B \varepsilon$

$$= Y^TY - \frac{1}{n}(Y^T1)^2$$
$$= (X\beta+\varepsilon)^T(X\beta+\varepsilon) - \frac{1}{n}(X\beta+\varepsilon)^T 1 1'(X\beta+\varepsilon)$$
$$= \beta^TX^TX\beta + 2\beta^TX^T\varepsilon + \varepsilon^T\varepsilon - \frac{1}{n}\beta^TX^T11^TX\beta - \dots + \varepsilon^T11'\varepsilon)$$

$$= \varepsilon^T\varepsilon - \frac{1}{n}\varepsilon^T11^T\varepsilon$$
$$= \varepsilon^T(I - \tfrac{1}{n}11^T)\varepsilon.$$

$$SSRes = \varepsilon^T(I - X(X^TX)^{-1}X^T)\varepsilon$$
$$\rightarrow SSTol = \varepsilon^T(I - \tfrac{1}{n}11^T)\varepsilon$$
$$SSReg = \varepsilon^T(X(X^TX)^{-1}X^T - \tfrac{1}{n}11')\varepsilon$$
$$Q_i = \varepsilon^T B^{(i)}\varepsilon.$$

$I - X(X^TX)^{-1}X^T = B^{(2)}$ SSReg

$X(X^TX)^{-1}X^T - \frac{1}{n}11' = B^{(3)}$ SSRes

$\frac{1}{n}11' = B^{(1)}$ ←

I

$Q_i := \varepsilon^T B^{(i)} \varepsilon$. $\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad I$

To show $\sum r_i = n$ when $r_i = r(B_i)$

note each $B^{(i)}$ is idempotent. rank = trace $\quad (tr(\frac{1}{n}11')$
$= tr(\frac{1}{n}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ \vdots & & & \vdots \\ 1 & \cdots & \cdots & 1 \end{bmatrix}))$

$r_2 = n - (p+1)$, $\quad r_1 = 1 \quad r_2 = p$

So $\sum r_i = n$.

$X \sim \chi^2_{d_1} \perp\!\!\!\perp Y \sim \chi^2_{d_2}$ then $\dfrac{X/d_1}{Y/d_2} \sim F_{d_1, d_2}$.